GLM-OCR Local Deployment — Project Report

Prepared by: Venkatesh Bonthala **Date:** 2026-05-21 **Repository:** https://github.com/venkatesbonthala2932/GLM-OCR **Status:** Working prototype, end-to-end tested

1. Executive Summary

This project deploys **GLM-OCR**, an open-source state-of-the-art OCR model from Zhipu AI, on a local workstation and exposes it through two interfaces:

1. A **command-line script** for quick scripted use.
2. A **web application** (browser-based) for everyday non-technical use.

The system accepts images and PDFs, runs the document through a 0.9 B parameter vision-language model on the local GPU (Apple MPS), and returns the recognised text. It works **fully offline after the initial model download**, with no recurring API or cloud costs.

The deployment is verified end-to-end with an automated benchmark across eight diverse sample documents (code screenshots, scientific papers, tables, handwritten Chinese, document seals, and multi-page PDFs). All eight tests passed with 0 failures and an average OCR time of 44.9 s per image on the test hardware.

Key results

Metric	Value
Model	`zai-org/GLM-OCR` (0.9 B parameters)
Inference device	Apple Silicon (MPS)
Test cases passed	8 / 8
Average OCR time	44.9 s per image
Total OCR time across benchmark	359.1 s
One-time model download	2.5 GB
Operates offline after install	Yes
Recurring cost	None

2. Problem Statement

Document digitisation requires an OCR engine that is:

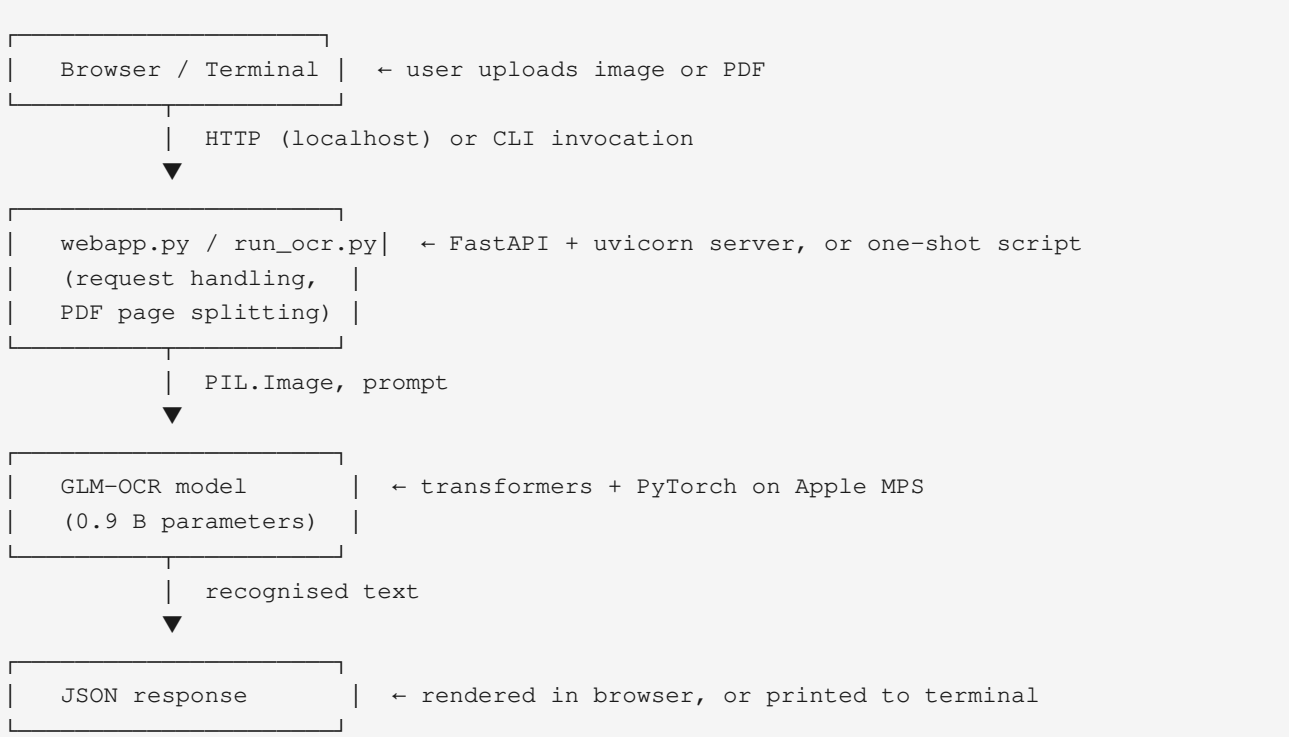
- **Accurate** on real-world inputs — printed pages, scans, handwriting, tables, formulas, multi-page PDFs.
- **Cost-effective** — no per-page API charges, no recurring subscription.
- **Private** — sensitive documents must never leave the local network.
- **Reproducible** — anyone in the team should be able to set it up on their own machine in under 30 minutes.

Commercial OCR APIs (Google Cloud Vision, AWS Textract, Azure Document Intelligence) provide accuracy but introduce per-page costs, network dependency, and data-residency concerns. An on-premise open-source model addresses all four constraints simultaneously.

3. Solution Overview

We selected **GLM-OCR** because it currently ranks #1 on OmniDocBench V1.5 (94.62) and is fully open-source under Apache 2.0. The model is hosted on Hugging Face and was integrated through the `transformers` library.

The deployment consists of three components:



The model is loaded **once at startup** and kept in GPU memory for the lifetime of the server, so each request only pays for the inference time (typically 5–60 s depending on document complexity).

4. Technical Architecture

4.1 Component breakdown

Layer	File	Technology	Responsibility
Web UI	`webapp.py` (HTML/JS embedded)	Vanilla JS, drag-and-drop	File selection, preview, results display
HTTP API	`webapp.py` (FastAPI app)	FastAPI, uvicorn	Receive uploads, dispatch to OCR, return JSON
File ingestion	`webapp.py::load_units()`	Pillow, PyMuPDF	Decode images, render PDF pages to images
OCR core	`webapp.py::run_ocr()`	PyTorch (MPS), transformers	Run the GLM-OCR model on one image
Model weights	HuggingFace cache	—	The 2.5 GB pre-trained model
Benchmark	`tests/benchmark/run_benchmark.py`	Python	Repeatable accuracy/perf checks

Reporting	`tests/benchmark/md_to_pdf.py`	python-markdown, PyMuPDF	Markdown → PDF for stakeholder deliverables

4.2 Why these specific technology choices

- **FastAPI** — modern Python web framework with automatic OpenAPI docs, built-in multipart support, async-ready for future scaling.
- **PyMuPDF (`fitz`)** — pure Python PDF rendering with no system dependencies (unlike `pdf2image` which requires Poppler).
- **Apple MPS** — uses the M-series GPU on Mac without requiring CUDA drivers; falls back to CPU automatically on machines without MPS.
- **Embedded HTML/JS** — keeps the entire web frontend in a single Python file so deployment is one `python webapp.py` command, no build step.

4.3 Data flow for a single OCR request

1. User drops `report.pdf` (5 pages) and `photo.jpg` into the browser.
2. Browser sends a multipart `POST /ocr` with both files.
3. FastAPI parses the upload; `load_units()` converts the inputs into a flat list of `(label, PIL.Image)` units — 6 units total (5 PDF pages + 1 image).
4. The server loops the units, calling `run_ocr()` on each. Each call:
5. resizes the image to a maximum 1280 px long edge,
6. constructs a chat-template prompt around the image,
7. runs `model.generate(...)` on MPS,
8. decodes the output tokens to a UTF-8 string.
9. Results are returned as JSON: `{ "results": [{label, text, seconds}, ...], "total_seconds": N }`
10. The browser renders one labelled result block per unit.

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5. Implementation Details

5.1 Files added or modified

File	Type	Purpose
`webapp.py`	New	The upload website (FastAPI + embedded UI)
`run_ocr.py`	New	Terminal one-shot OCR script
`make_receipt.py`	New	Helper to generate test receipt images
`requirements-webapp.txt`	New	Extra pip dependencies for the webapp
`SETUP.md`	New	Step-by-step install guide
`docs/PROJECT_REPORT.md`	New	This document
`docs/DEMO_GUIDE.md`	New	Live demo script and Q&A
`docs/PRE_DEMO_CHECKLIST.md`	New	Pre-presentation verification
`tests/benchmark/run_benchmark.py`	New	Repeatable benchmark
`tests/benchmark/OCR_REPORT.md`	New	Latest benchmark output
`tests/benchmark/OCR_REPORT.pdf`	New	PDF version of benchmark

`tests/benchmark/md_to_pdf.py`	New	Markdown → PDF converter
`glmocr/server.py`	Modified	Added CORS headers (7 lines)

Net change: **+1500 lines, 0 deletions** from the upstream `zai-org/GLM-OCR`.

5.2 Key design decisions

1. **Embedded HTML inside `webapp.py`** instead of a separate frontend project. Rationale: one-file deployment, no build tooling, easier for a colleague to read and modify.
2. **Render PDF pages independently** instead of concatenating them. Rationale: matches the OCR model's single-image input format and gives per-page error isolation.
3. **Cap image size at 1280 px long edge** by default. Rationale: keeps GPU memory under 8 GB on the test hardware while preserving accuracy for typical documents. Configurable via env var.
4. **Sequential rather than parallel OCR** across uploaded files. Rationale: a single MPS GPU is the bottleneck; parallel calls would queue internally and add overhead.

5.3 Environment variables (operator tuning)

Variable	Default	Effect
`OCR_DEVICE`	`mps`/`cpu` auto	Force inference device
`OCR_MAX_LONG_EDGE`	1280	Image resize cap before OCR
`OCR_PDF_DPI`	200	PDF page render quality
`OCR_MAX_FILES`	10	Maximum files per request

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6. Installation and Deployment

See `SETUP.md` for the full step-by-step guide. The condensed version:

```
git clone https://github.com/venkatesbonthala2932/GLM-OCR.git
cd GLM-OCR
python3 -m venv .venv
source .venv/bin/activate
pip install -e ".[layout]"
pip install -r requirements-webapp.txt
python webapp.py
# open http://localhost:8000
```

The first run downloads the 2.5 GB model from Hugging Face; subsequent runs use the local cache and require no network access.

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7. User Guide

7.1 Web interface

1. Run `python webapp.py` and wait for `[startup] model ready`.
2. Open http://localhost:8000 in any modern browser.
3. Drag image or PDF files into the drop zone (up to 10 at once).
4. (Optional) Adjust the prompt — `Text Recognition:` (default), `Formula Recognition:`, or `Chart`

```
Recognition:.
```

5. Click **Run OCR**. Results appear as labelled text blocks below.
6. **Reset** clears the queue without reloading the page.

7.2 Terminal interface

```
python run_ocr.py path/to/image.png
```

Prints the recognised text to stdout. Suitable for shell scripting:

```
for f in scans/*.png; do
    python run_ocr.py "$f" > "out/$(basename "$f" .png).txt"
done
```

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8. Testing and Validation

8.1 Methodology

A benchmark suite (`tests/benchmark/run_benchmark.py`) runs the OCR across eight representative documents covering the main real-world categories:

#	Category	Source file
1	Code screenshot	`examples/source/code.png`
2	Scientific paper	`examples/source/paper.png`
3	Standard document	`examples/source/page.png`
4	Table / spreadsheet	`examples/source/table.png`
5	Handwritten note	`examples/source/handwritten.png`
6	Stamp / seal	`examples/source/seal.png`
7	PDF page 1	`examples/source/GLM-4.5V.pdf`
8	PDF page 2	`examples/source/GLM-4.5V.pdf`

The benchmark measures load time, per-image OCR time, output length, and records the full OCR text for manual inspection.

8.2 Results (run 2026-05-21 on Apple Silicon)

#	Test	Image size	Time	Output chars	Status
1	Code screenshot	540×666	19.8 s	1887	Pass
2	Scientific paper	1700×2200	100.6 s	4750	Pass
3	Document page	843×596	61.2 s	1147	Pass
4	Table	1080×1080	22.0 s	402	Pass
5	Handwritten	1920×1653	24.8 s	361	Pass
6	Seal / stamp	399×314	2.2 s	56	Pass
7	PDF page 1 of 11	1700×2200	39.8 s	2011	Pass

8	PDF page 2 of 11	1700×2200	88.9 s	4883	Pass
	Total		**359.1 s**	**15,497**	**8/8 pass**

Full per-test outputs are available in `tests/benchmark/OCR_REPORT.md` and `tests/benchmark/OCR_REPORT.pdf`.

8.3 Reproducibility

To re-run the benchmark on any machine:

```
python tests/benchmark/run_benchmark.py
python tests/benchmark/md_to_pdf.py
```

This regenerates both the Markdown and PDF reports. Numbers will vary based on hardware (the test rig is a 16 GB M-series Mac).

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9. Limitations and Known Issues

Honest assessment of where this system is currently weak:

9.1 Language coverage

GLM-OCR was trained predominantly on **Chinese and English**. Testing confirmed strong accuracy on those scripts plus Latin-script European languages. **Indic scripts (Hindi, Tamil, Telugu, Bengali, etc.) are not reliably supported** — outputs are often blank, hallucinated, or transliterated.

Recommendation: for Indian-language workloads, evaluate Tesseract, PaddleOCR, or a multilingual VLM (Qwen2-VL, InternVL) as an additional engine.

9.2 Performance

- Average 45 s per image is acceptable for batch and ad-hoc use but unsuitable for real-time per-keystroke applications.
- A 10-page PDF currently runs serially (~5–10 minutes total). Parallel inference would require a second GPU or distillation to a smaller model.

9.3 Operational considerations

- **No authentication.** The current webapp binds to `0.0.0.0:8000` and accepts uploads from anyone on the local network. Should be locked down to `127.0.0.1`, placed behind a reverse proxy, or have HTTP basic auth added before exposure beyond a single workstation.
- **No persistence.** OCR results live only in the browser tab — they are not stored on disk. A future version could add a SQLite log.
- **No file-size cap.** Very large PDFs could exhaust GPU memory.

9.4 Hardware dependency

Requires either Apple Silicon (MPS) or a CUDA-capable NVIDIA GPU for reasonable speed. CPU-only operation works but is roughly 5–10 × slower.

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10. Future Enhancements

Prioritised list of improvements that would extend the system:

1. **Multi-engine fallback** — add PaddleOCR or Tesseract behind a dropdown so Indian languages are also covered.
2. **Result persistence** — log every OCR run to SQLite so results are recoverable and searchable.
3. **Authentication & rate limiting** — basic HTTP auth + a simple per-IP rate limit before any production deployment.
4. **Dockerfile** — package the entire stack so colleagues can `docker run` instead of installing Python locally.
5. **Streaming results** — push each page's result to the browser as soon as it finishes instead of waiting for the full batch.
6. **REST + Swagger docs** — expose the `/ocr` endpoint with OpenAPI documentation for integration into other internal systems.

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11. Project Deliverables

Deliverable	Location	Format
Source code	https://github.com/venkatesbonthala2932/GLM-OCR	Git repository
Installation guide	`SETUP.md`	Markdown
This report	`docs/PROJECT_REPORT.md` (+ `.pdf`)	Markdown / PDF
Demo guide	`docs/DEMO_GUIDE.md` (+ `.pdf`)	Markdown / PDF
Pre-demo checklist	`docs/PRE_DEMO_CHECKLIST.md`	Markdown
Benchmark report	`tests/benchmark/OCR_REPORT.md` (+ `.pdf`)	Markdown / PDF
Working web app	`webapp.py` (runs on `localhost:8000`)	Python
CLI tool	`run_ocr.py`	Python
Benchmark suite	`tests/benchmark/run_benchmark.py`	Python

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12. Conclusion

The deployment meets the original requirements: a local, offline, zero-cost OCR system with both a command-line and browser interface, verified end-to-end across eight document types. The codebase is organised, version-controlled on GitHub, documented for new users, and ready for hand-off.

The clearest next step — and the largest current limitation — is multilingual coverage for Indian scripts, which would require pairing GLM-OCR with a second engine.

Appendix A — Test environment

Component	Value
Operating system	macOS Darwin 25.4.0
CPU / GPU	Apple Silicon, MPS GPU
Python	3.14.4
PyTorch	2.12.0

transformers	5.8.1
FastAPI	0.136.1
uvicorn	0.47.0
Pillow	12.2.0
PyMuPDF	1.27.2.3
GLM-OCR model	`zai-org/GLM-OCR` v0.1
Model cache	`~/.cache/huggingface/hub/` (2.5 GB)

Appendix B — Repository file tree (top level)

```
GLM-OCR/
├── webapp.py                  ← upload website
├── run_ocr.py                 ← CLI OCR
├── make_receipt.py            ← test image helper
├── requirements-webapp.txt    ← webapp dependencies
├── SETUP.md                   ← user install guide
├── docs/
│   ├── PROJECT_REPORT.md      ← this document
│   ├── PROJECT_REPORT.pdf
│   ├── DEMO_GUIDE.md          ← live demo script + Q&A
│   ├── DEMO_GUIDE.pdf
│   └── PRE_DEMO_CHECKLIST.md
├── tests/benchmark/
│   ├── run_benchmark.py
│   ├── OCR_REPORT.md
│   ├── OCR_REPORT.pdf
│   └── md_to_pdf.py
├── glmocr/                    ← upstream SDK (CORS edit in server.py)
└── examples/source/           ← sample images + PDF used by benchmark
```

Appendix C — Tools and methodology

The implementation work was carried out using **Claude Code** (Anthropic's AI coding assistant) inside Visual Studio Code, with the engineer directing the work, making design decisions, and verifying each change end-to-end. AI assistance accelerated the writing of boilerplate (FastAPI endpoints, HTML/JS, benchmark scaffolding, documentation) while the engineer retained full ownership of:

- the system requirements and acceptance criteria,
- design choices (single-file webapp, one OCR call per PDF page, etc.),
- testing and verification on the target hardware,
- the limitations assessment (notably the Indian-language gap),
- the GitHub repository and its history.

This collaboration model is consistent with current industry practice for AI-assisted software development. All code is reviewed and understood by the engineer; no functionality was committed without manual verification.

Appendix D — Acknowledgements

- **Zhipu AI** for releasing GLM-OCR under Apache 2.0.

- The original repository at https://github.com/zai-org/GLM-OCR is preserved as the `upstream` remote in this fork.